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SUMMARY

Software Development Engineer with ~4 years building distributed systems, AI/ML pipelines, and large language models. At IBM Software Labs, architected microservices serving ~1,000 engineers at 4–5K req/min and shipped production AI auto-healing. Open-source contributor with hands-on LLM pretraining (Phoenix 125M), QLoRA fine-tuning (87% SQL accuracy), RAG pipelines, agentic systems, and MCP server development.

EXPERIENCE

Software Development Engineer 2

Jul 2024 – Present

IBM Software Labs

Bengaluru, India

- Re-architected IBM Software Agile Tester from a monolith into 4 distributed microservices, serving ~1,000 engineers with 4–5K req/min at sub-second latency
- Designed and shipped AI auto-healing using a 3-tier cascade (CSS selectors, text embeddings, IBM Granite 3.3 VLM) that self-corrects broken UI locators across 5K+ test cases; 100% accuracy on Firefox, 83% on Chrome; saves ~20 engineer-hours/week
- Led Java 8 to 17 migration across enterprise microservices, improving runtime performance by 30% and reducing GC overhead
- Built CI/CD pipelines with Jenkins, Docker, and NGINX, achieving 30% faster builds and 99% uptime SLA; mentors junior engineers on system design and code quality

Software Engineer 2

Nov 2023 – Jul 2024

Software AG (now IBM)

Bengaluru, India

- Extended company-wide AI Chatbot with LLM integration and full RAG pipeline (FAISS, Milvus, Knowledge Graph, auto ticket creation), achieving 70% reduction in support tickets (~200 → ~60/week)
- Built Failure Prediction system using PyTorch and time-series analysis, targeting 99.99% system availability
- Built Semantic Search Engine using NLP, Knowledge Graphs, and FAISS – won TechInterrupt Hackathon (1st in India, 4th internationally)

Software Engineer

Aug 2022 – Nov 2023

Software AG

Bengaluru, India

- Developed enterprise integration platform features (webMethods) using Java, Spring Boot, and RESTful APIs; built microservices for the API Management platform
- Recognized as 2023 Star Performer; received Pace Setter (Oct 2023) and Special Contribution (May 2023) awards

PROJECTS

Phoenix 125M + Sweta-Hi/Kn | *Open-Source LLMs* | *HuggingFace*

2025 – 2026

- Pretrained a 125M-param decoder-only LLM from scratch on a single RTX 3080 Ti; LLaMA-style architecture (RoPE, SwiGLU, RMSNorm); custom BPE tokenizer, data pipeline, and training loop with mixed-precision and gradient checkpointing
- Trained on ~2B tokens; WinoGrande 0.507 – matches GPT-Neo 125M & OPT-125M trained on 150–300× more data; Apache 2.0. Also built Sweta-Hi (Hindi) and Sweta-Kn (Kannada) 125M models with custom Devanagari/Kannada tokenizers

Mistral-7B Text-to-SQL | *QLoRA Fine-Tune* | *Apache 2.0*

2026

- Fine-tuned Mistral 7B for Text-to-SQL via QLoRA (4-bit NF4, r=64, $\alpha=16$); LoRA adapter merged into base weights; served via vLLM
- Spider benchmark (500 examples): Exact Set Match 87.0% (+77.8 pp over zero-shot baseline); Valid SQL 97.4% (+88.6 pp)

ATIS | *Autonomous Trading Strategy Research System*

2025 – Present

- Agentic research loop: Playwright web research → Backtrader backtesting (slippage, STT, impact cost) → Optuna hyperparameter optimisation; strategy store in Qdrant prevents duplicate generation
- Self-building knowledge graph maps cross-asset dependencies; 98 strategies evaluated; 19.4% quality-gate pass rate (Sharpe > 2, Win Factor > 1, out-of-sample validated)

LinkedIn Post Swarm | *Multi-Agent Content Pipeline*

2025

- Sequential agent swarm (Research → Critic → SEO) orchestrated by frontier LLM; ~80% of token load handled by local SLMs (Ollama), cutting per-post cost by 85% (INR 83 → INR 10–20)
- Pipeline drove 5× LinkedIn audience growth (400 → 2,000 followers) across 11 posts; HITL approval gate via Telegram before publish

Dhan MCP Server | *Open-Source* | [GitHub](#)

2025

- MCP server for DhanHQ trading API exposing live OHLC, options chain with Greeks ($\Delta/\Theta/\Gamma/\text{Vega}$), historical OHLCV, instrument lookup, and order management
- Rate limiting, sandbox mode (DHAN.FAKE=1), and 3 config methods (env vars, config.ini, flag); MIT licensed, Python

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, Go

Backend: Spring Boot, Flask, FastAPI, Node.js, REST APIs, Microservices, Apache Kafka, gRPC

AI/ML: PyTorch, Transformers, LLM Pretraining, QLoRA/LoRA/PEFT, RAG, LangChain, LangGraph, vLLM, HuggingFace, FAISS, Milvus, Qdrant, NLP, Knowledge Graphs, Optuna

Databases: MySQL, MongoDB, Cassandra, Redis, SQLite

DevOps & Tools: Docker, Kubernetes, Jenkins, CI/CD, NGINX, Git, Linux

Other: MCP (Model Context Protocol), Playwright, Backtrader

EDUCATION

Bangalore Institute of Technology

2018 – 2022

Bachelor of Engineering, Computer Science

CERTIFICATIONS

Enterprise Design Thinking – IBM | Apache Kafka – IBM | Machine Learning Specialization – deeplearning.ai | Neural Networks and Deep Learning – deeplearning.ai | Improving Deep Neural Networks – deeplearning.ai